THE PERIODIC TABLE

COURSE CONTENT

1. History of the atom
2. Sub-atomic Particles
   - protons, electrons and neutrons
3. Atomic number and Mass number
4. Isotopes and Ions
5. Periodic Table
   - Groups and Periods
6. Properties of metals and non-metals
7. Metalloids and Alloys

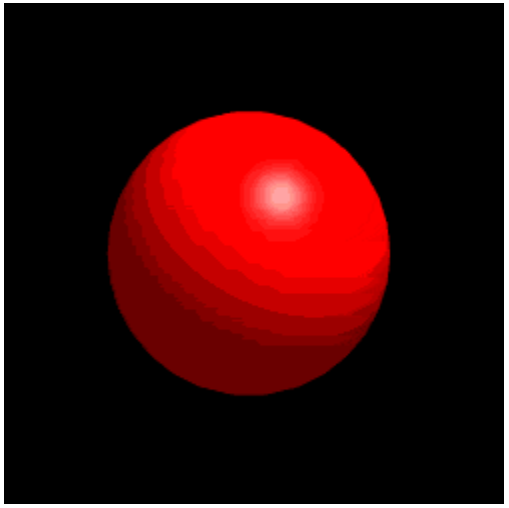
OBJECTIVES

- Describe an atom in terms of the sub-atomic particles
- Identify the location of the sub-atomic particles in an atom
- Identify and write symbols of elements (atomic and mass number)
- Explain ions and isotopes
- Describe the periodic table
  - Major groups and regions
  - Identify elements and describe their properties
- Distinguish between metals, non-metals, metalloids and alloys

Atom Overview

- The Greek philosopher Democritus (460 B.C. – 370 B.C.) was among the first to suggest the existence of atoms (from the Greek word “atomos” = uncuttable)
  - He believed that atoms were ***indivisible*** and ***indestructible***
  - His ideas did agree with later scientific theory, but did not explain chemical behavior, and was *not based on the scientific method* – but just philosophy

John Dalton(1766-1844)

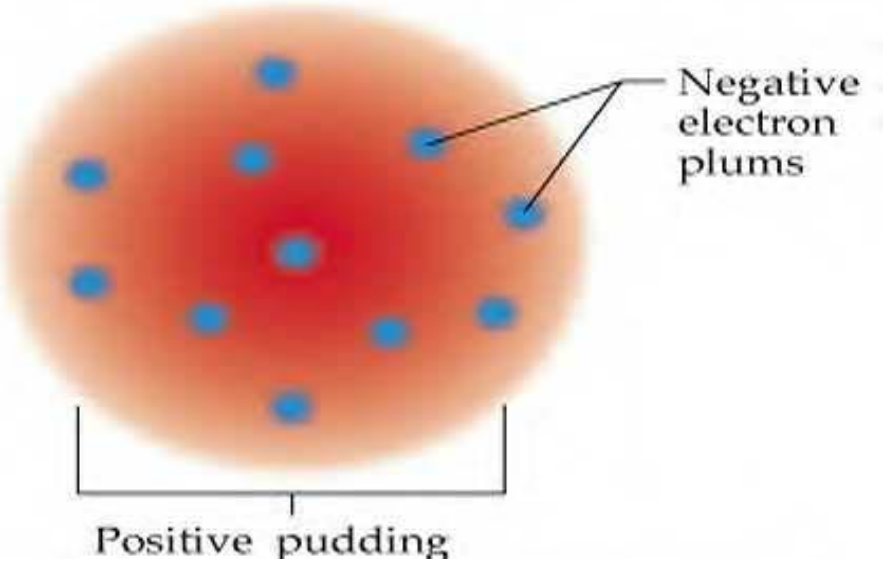


In 1803, he proposed :

1. All matter is composed of atoms.
2. Atoms cannot be created or destroyed.
3. All the atoms of an element are identical.
4. The atoms of different elements are different.
5. When chemical reactions take place, atoms of different elements join together to form compounds.

J.J.Thomson (1856-1940)

1. Proposed the first model of the atom.
2. 1897- Thomson discovered the electron (negatively- charged) – cathode rays
3. Thomson suggested that an atom is a positively- charged sphere with electrons embedded in it.



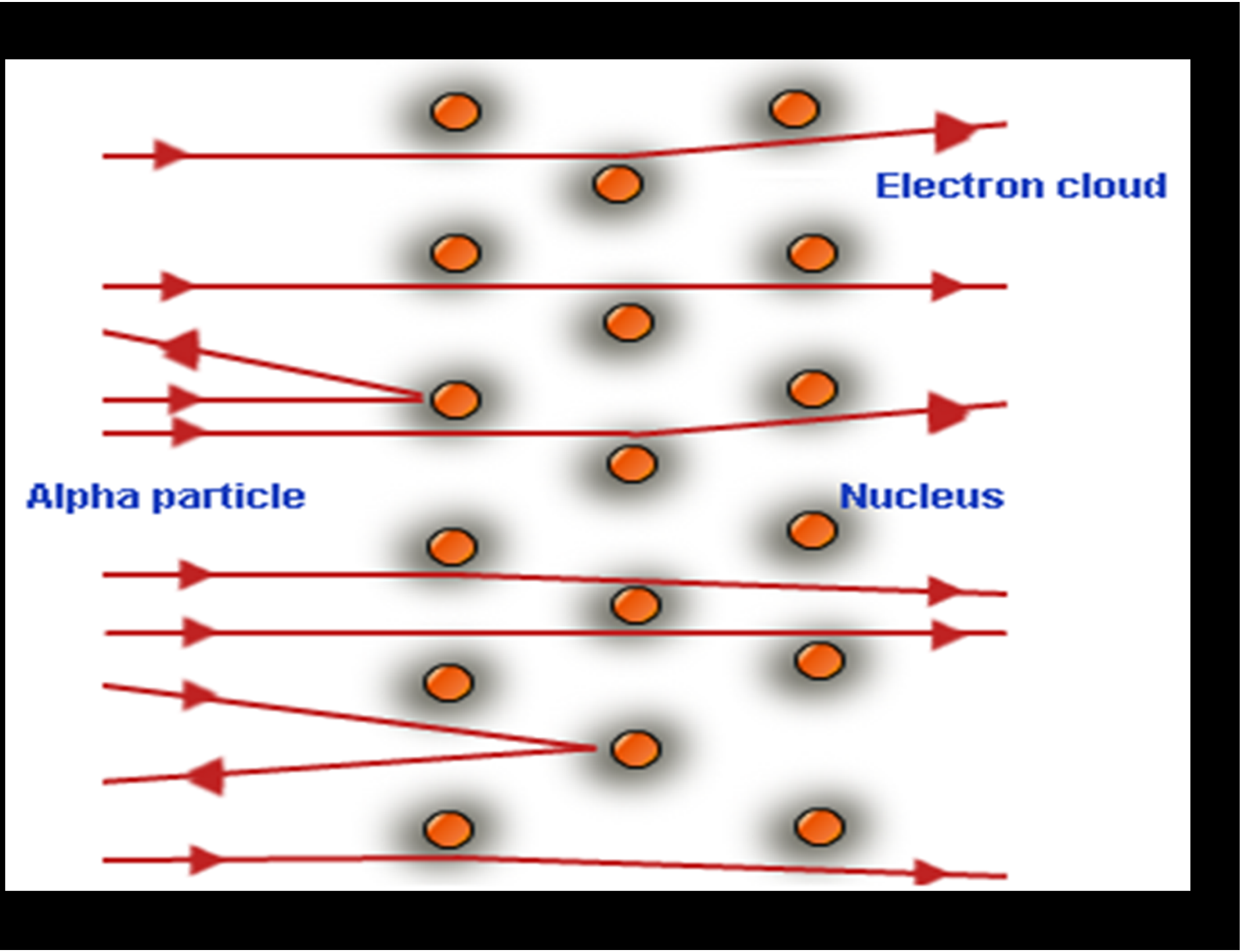
Ernest Rutherford (1871-1937)

1. 1914- Rutherford discovered the proton
2. Rutherford model was based on the alpha particle scattering experiment
3. He proposed

   1) all the positive charge of an atom is concentrated in the nucleus

   2) an atom consists of a positively-charged nucleus with a cloud of electrons surrounding the nucleus




Electron cloud
Alpha particle
Nucleus

Neils Bohr (1885-1962)

- He was a student of Rutherford
- He proposed

  1) electrons are arranged in orbits (electron shells) around the nucleus of the atom

  2) electrons move in a particular path, have a fixed energy.
- To move from one orbit to another, an electron must gain or lose the right amount of energy

Atom Overview

- **ATOM**- is the smallest particle of an element that maintains the characteristics of that element
- **ELEMENT**- is a pure substance that cannot be split up into 2 or more simpler substances by chemical processes
- Atom is made up:
  - Nucleus
  - Electron cloud or shells
- Nucleus and electron cloud consists of 3 sub-atomic particles:
  - Protons
  - Neutrons
  - Electrons

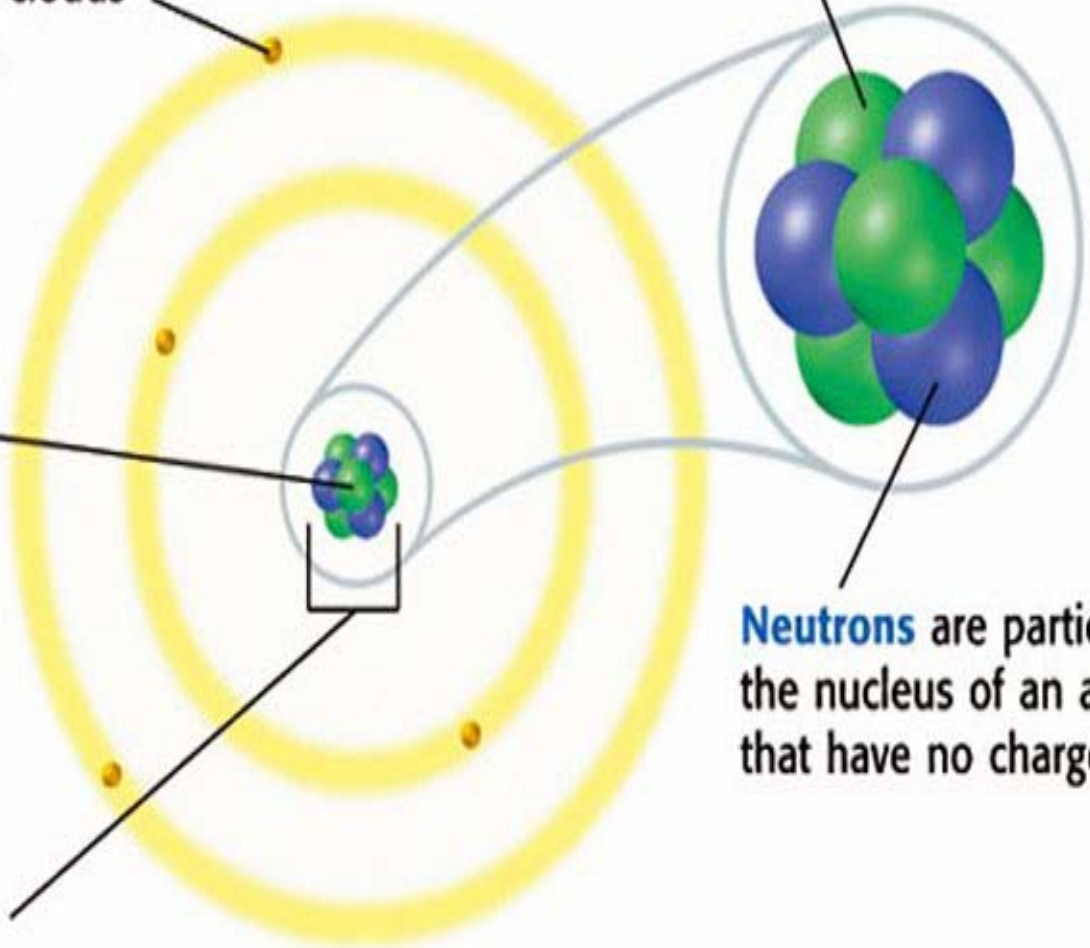
What is an atom made of?

- **The Nucleus**
  - **Protons**
    - ***Positively charged*** particles in the nucleus
  - **Neutrons**
    - Particles of the nucleus that have ***no electrical charge***
- **Electron cloud/ shells**
  - **Electrons**
    - ***Negatively charged*** particles in atoms
    - Found around the nucleus within ***electron clouds***

Parts of an Atom

Electrons are negatively charged particles found in electron clouds outside the nucleus. The size of the electron clouds determines the size of the atom.

Protons are positively charged particles in the nucleus of an atom.



The **nucleus** is the small, dense, positively charged center of the atom. It contains most of the atom's mass.

Neutrons are particles in the nucleus of an atom that have no charge.

The diameter of the nucleus is 1/100,000 the diameter of the atom.

Sub-atomic Particles

Particle	Charge	Mass (g)	Location
Electron (e^-)	-1	9.11×10^{-28}	Electron cloud
Proton (p^+)	+1	1.67×10^{-24}	Nucleus
Neutron (n^o)	0	1.67×10^{-24}	Nucleus

Atomic Number

- Atoms are composed of ***identical*** protons, neutrons, and electrons
  - How then are atoms of one element different from another element?
- Elements are different because they contain different numbers of PROTONS
- The atomic number of an element is the number of protons in the nucleus
- # protons in an atom = # electrons

Atomic Number

Atomic number (Z) of an element is the number of protons in the nucleus of each atom of that element.

Element	# of protons	Atomic # (Z)
Carbon	6	6
Phosphorus	15	15
Gold	79	79

Mass Number

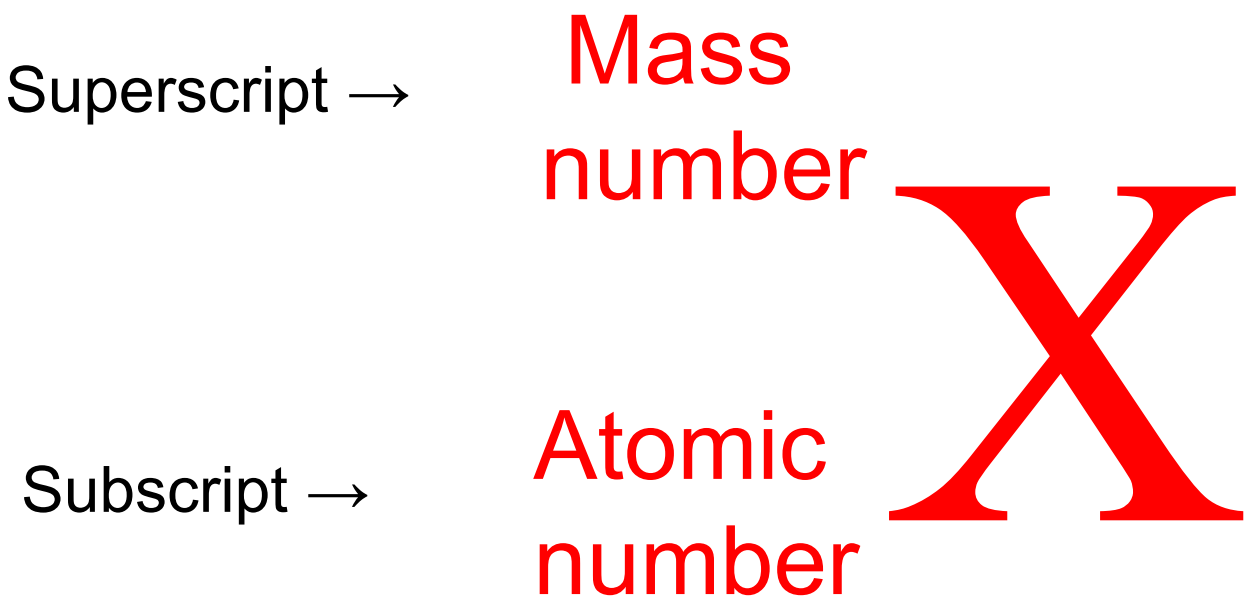
Mass number (A) is the number of protons and neutrons in the nucleus of an element:

Mass # = p^+ + n^0

Nucleus	p^+	n^0	e^-	Mass #
Oxygen - 18	8	10	8	18
Arsenic - 75	33	42	33	75
Phosphorus - 31	15	16	15	31

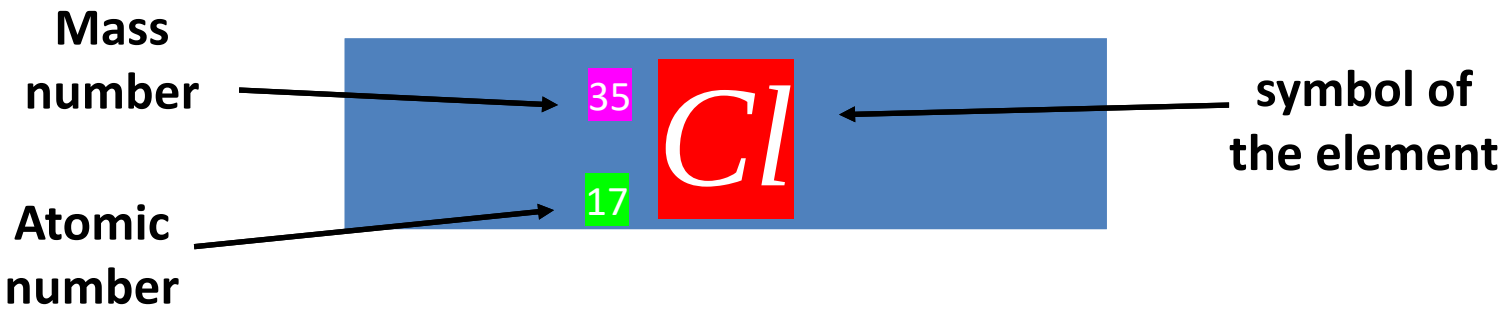
Complete Symbols

- Contain the symbol of the element, the mass number and the atomic number.



Symbol Form

- written in symbol form



- symbol tells us that the atom
  - is of element chlorine
  - has atomic number 17 (so it contains 17 protons)
  - has 17 electrons (number of protons = number of electrons)
  - has mass number 35 (so number of protons + number of neutrons = 35)
  - it must contain 35 – 17 = 18 neutrons

Symbols

- Find each of these:

  a) number of protons

  b) number of neutrons

  c) number of electrons

  d) Atomic number

  e) Mass Number

$$^{80}_{35}\mathrm{Br}$$

Symbols

- If an element has an atomic number of 34 and a mass number of 78, what is the:

  a) number of protons

  b) number of neutrons

  c) number of electrons

  d) complete symbol

Isotopes

- Atoms of an element that have the same number of protons and electrons ,but different numbers of neutrons.

❖**THEREFORE,DIFFERENT MASS NUMBER!!!!**

– Hydrogen isotopes

- Hydrogen has 1 proton, 1 electron and 0 neutrons
- Deuterium has 1 proton, 1 electron and 1 neutron
  - Therefore, it is heavier than hydrogen but has similar chemical properties and slightly different physical properties
- Tritium has 1 proton, 1 electron and 2 neutrons

Isotopes	$^{23}_{11}Na$	$^{24}_{11}Na$
number of protons	11	11
number of electrons	11	11
number of neutrons	23 - 11 = 12	24 – 11 = 13

Ions

- **Ion** – electrically charged particle
- Thus, atoms whose number of electrons does NOT EQUAL the number of protons
- Either positively or negatively charged
- Positively charged = loses one or more electrons
- Negatively charged = gains one or more electrons
- Determining the charge of an ion:

  Overall charge = # of protons - # of electrons

Symbol for Ions

- Includes:
  - Atomic number
  - Mass number
  - Element symbol
  - Charge
- Example:
  - 35 protons, 45 neutrons, 36 electrons (gained 1 electron)

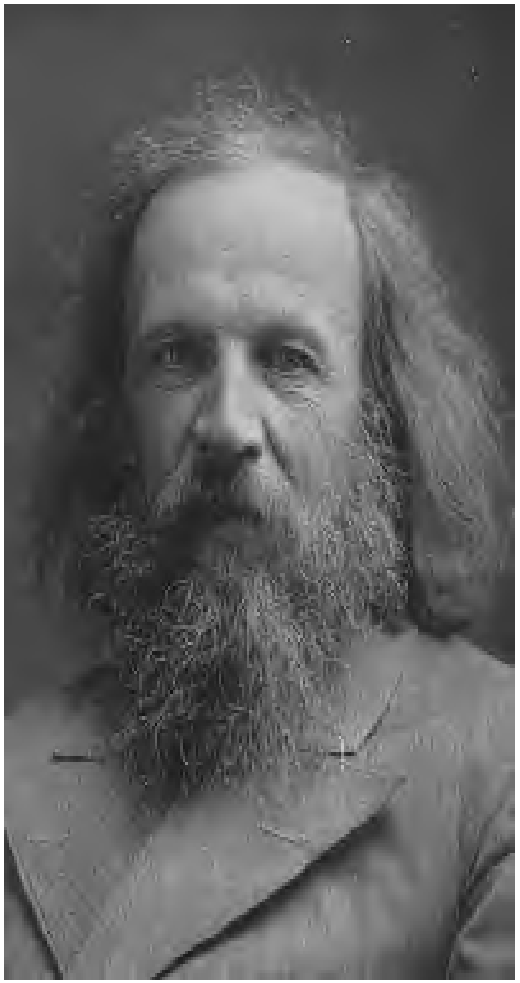
$$^{80}_{35}\mathrm{Br}^{-1}$$

  - 12 protons, 12 neutrons, 10 electrons (lost 2 electrons)

$$^{24}_{12}\mathrm{Mg}^{+2}$$

The Periodic Table

- In 1869,Dmitri Ivanovitch Mendeléev created the first accepted version of the periodic table.
- He grouped elements according to their atomic mass, and as he did, he found that the groups had similar chemical properties.
- Blank spaces were left open to add the new elements he predicted would occur.



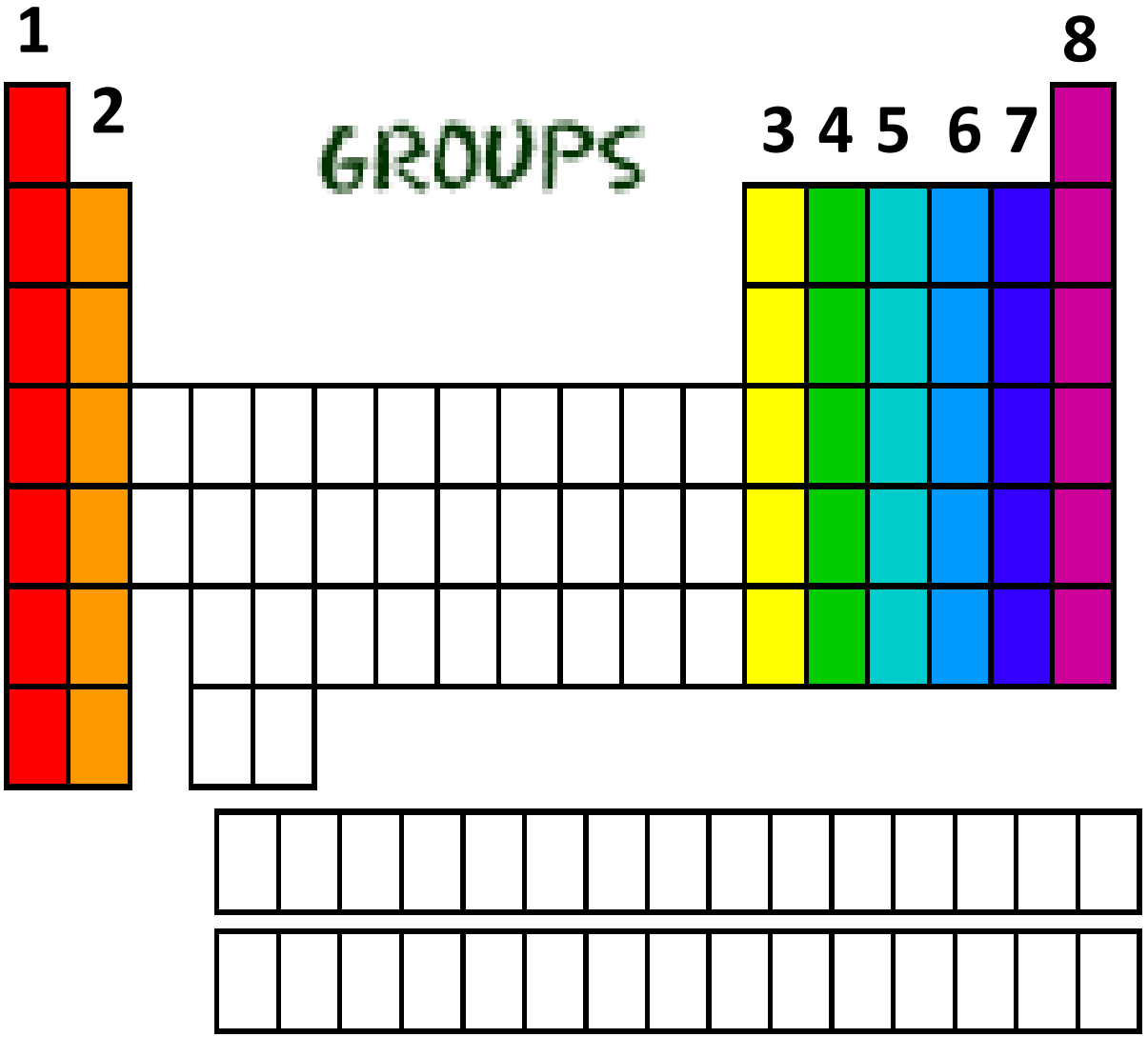
The Periodic Table

- Periodic Law - When the elements are arranged in order of increasing atomic number, there is a periodic repetition of their physical and chemical properties
- Modern Periodic consists of:
  - **Periods** - Horizontal rows of the periodic table (side to side)
  - **Groups or families** - – vertical (up and down) column of elements in the periodic table

The Periodic Table

- Groups – Arranged by the number of valence electrons i.e. number of electrons in the outer shell
  - Elements in the same group has same number of valence electrons
  - Each group has similar chemical and bonding properties
  - 8 groups
- Periods – Arranged by increasing atomic number
  - Elements in a period have the same number of electronic shells
  - 7 periods

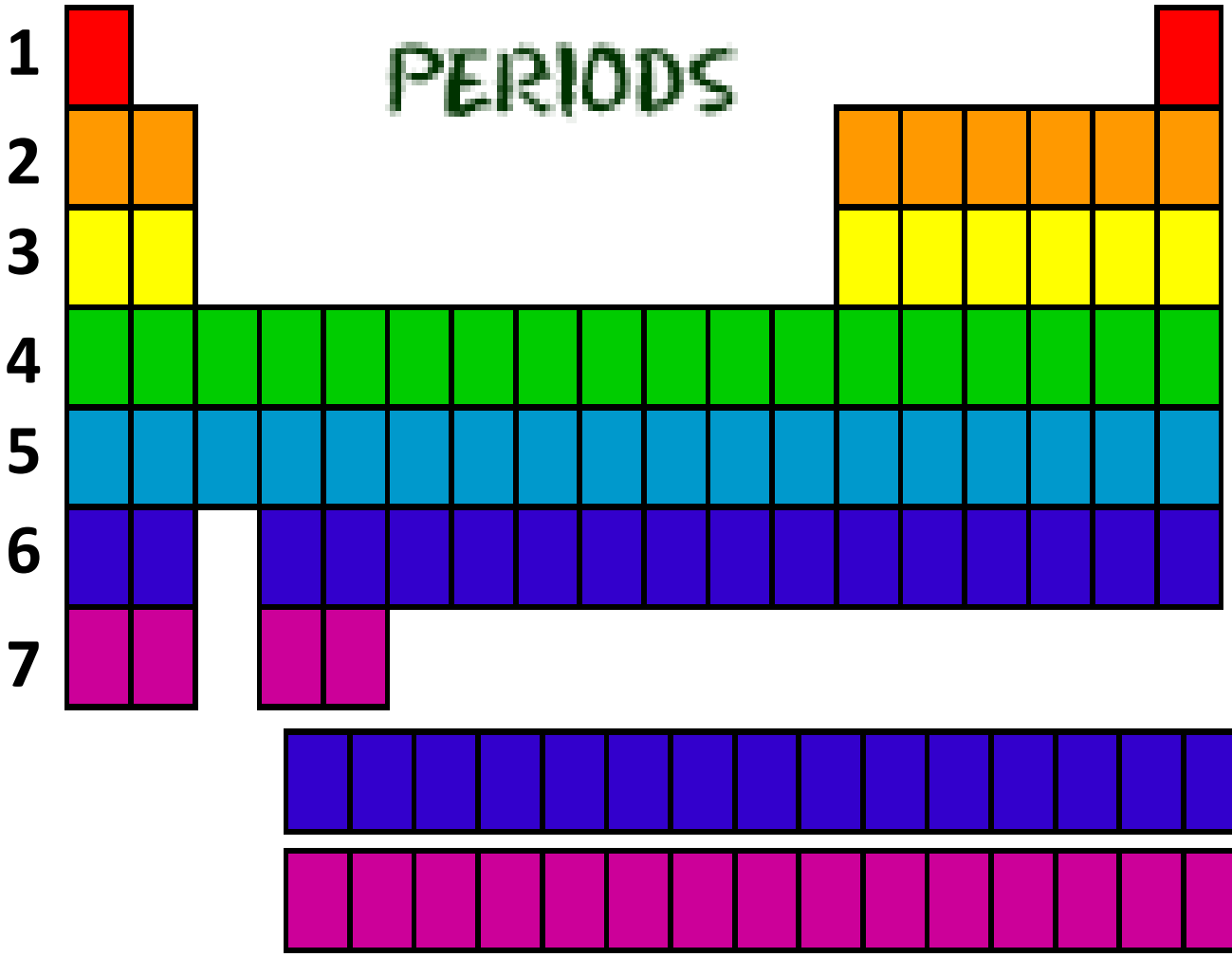
Periodic Table



Groups go down on the periodic table

Elements in the same group, have the same number of valence electrons

Periodic Table

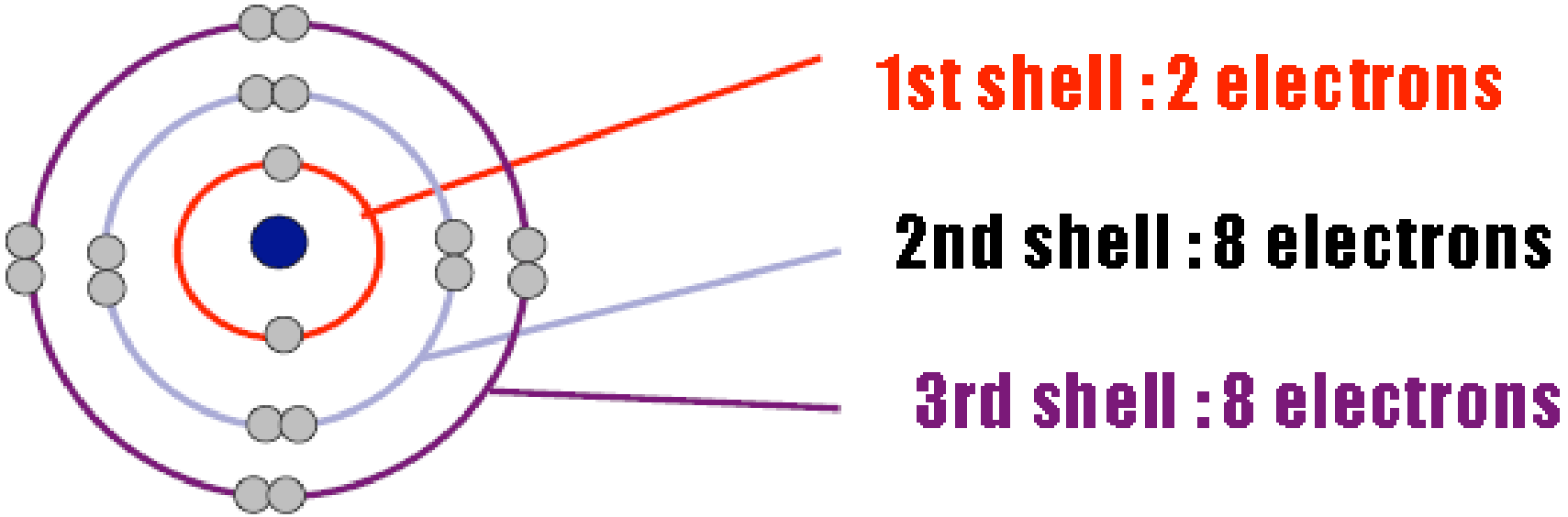


Periods go across on the periodic table

Periods have the same number of “shells”

Arrangement of electrons in the atom

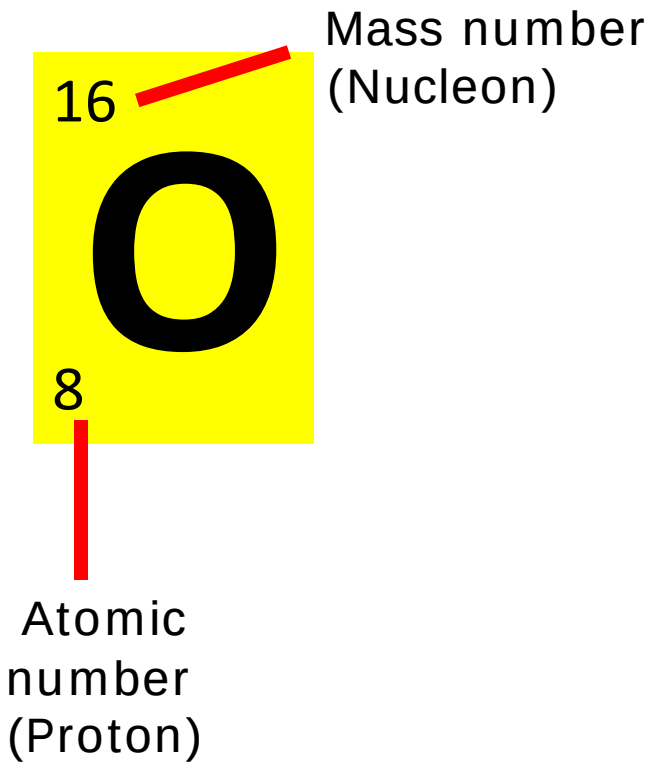
- Electrons are arranged in energy levels i.e. Electron shells, around the nucleus
- Each energy level can only hold a certain # of electrons

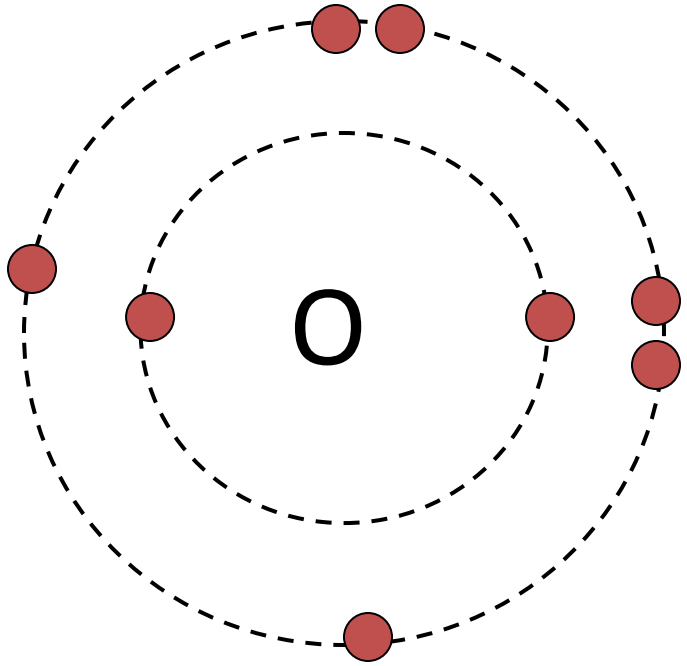


Arrangement of electrons in the atom

- Main rule – electrons always go into the shell nearest to the nucleus, if there is room. Once the shell is filled up, the electrons go into the next available shell.
- Outermost shell of an atom is called the valence shell
- This shell should have electrons before it can be called a valence shell
- The electrons in the valence shell are called the valence electrons

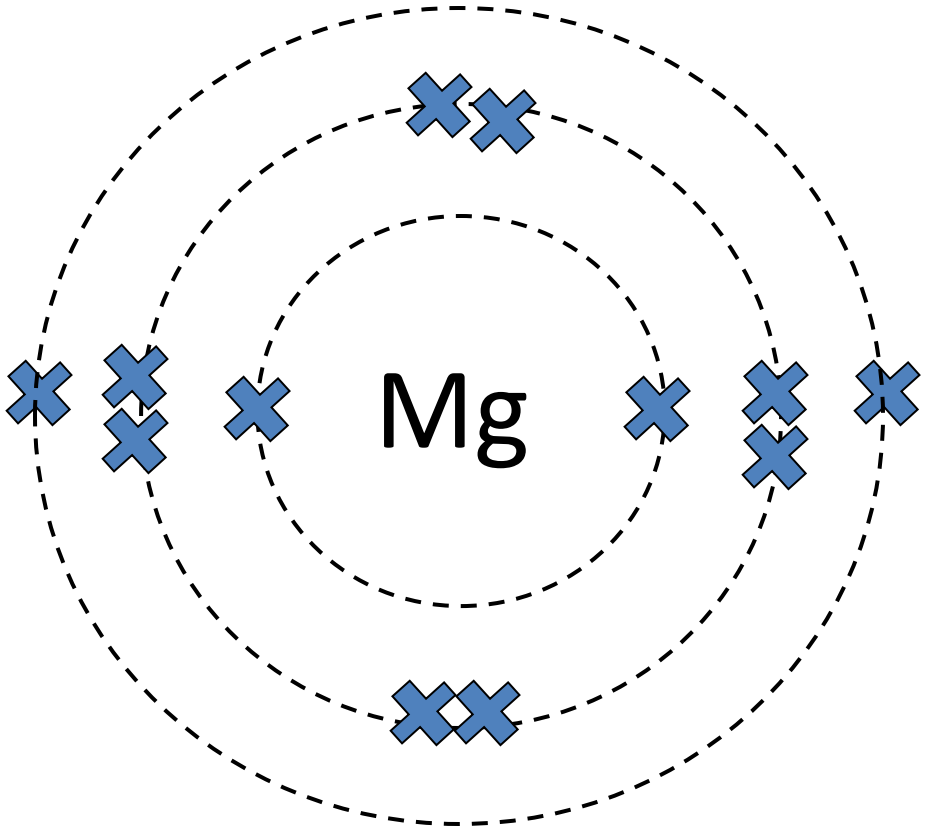
The Electronic Configuration of Atoms





The Electronic Configuration of Atoms





Groups

- Columns of elements are called groups or families.
- Elements in each group have similar but not identical properties.
- All elements in a group have the same number of valence electrons.
- Include:
  - Group A: Alkali metals, alkali earth metals, boron, carbon, nitrogen, oxygen, halogens and noble/inert gases
  - Group B: transition metals

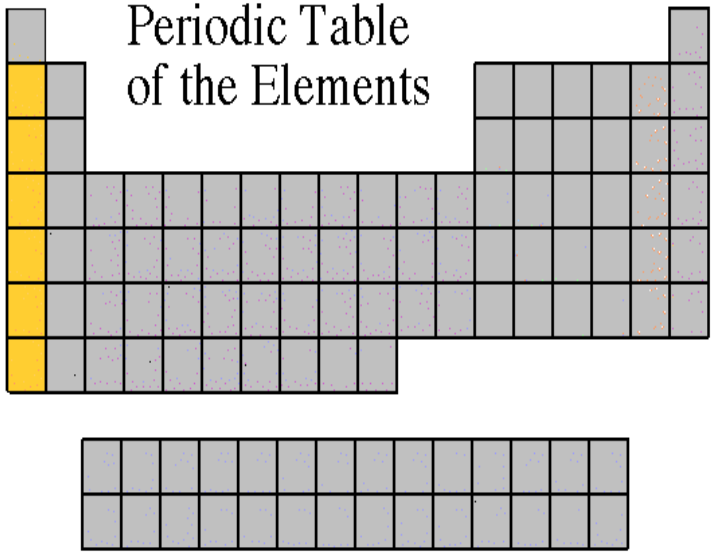
    lanthanides and actinides (inner transition metals)

Hydrogen

- The hydrogen found on top group I, but it is not a member of that group
- Hydrogen is in a class of its own
- It’s a colourless gas at room temp
- Diatomic, reactive gas
- It has one proton and one electron in its one and only energy level.
- Promising alternative fuel source

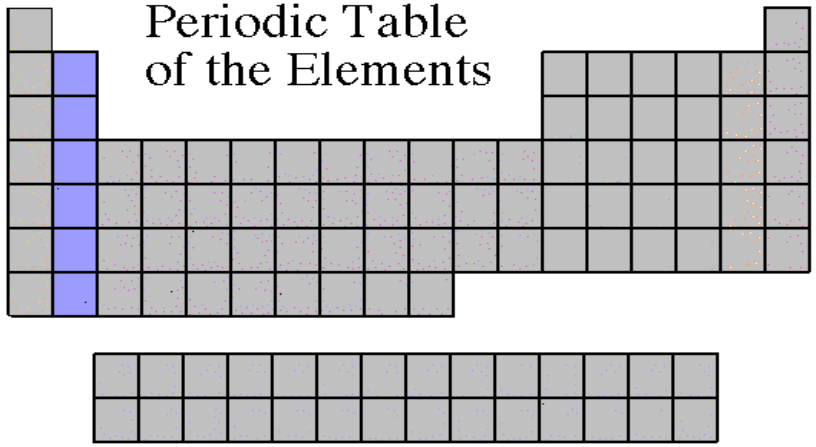
Alkali Metals

- The alkali family is found in the first column of the periodic table.
- Atoms of the alkali metals have a single electron in their outermost level, in other words, 1 valence electron.
- Tend to lose 1 electron (form + ions)
- Alkali metals are never found as free elements in nature. They are always bonded with another element.
- They are shiny, have the consistency of clay, and are easily cut with a knife.
- Highly reactive, stored under oil
- Density less than water
- mp and bp are very low compared to other metals



Alkaline Earth Metals

- 2 valence electrons
- Tend to lose the 2 electrons (form +2 ions)
- They are always combined with non-metals in nature e.g. metal oxides
- Alkaline earth metals include magnesium (Mg) and calcium (Ca), which important mineral nutrient, among others.



Boron Family

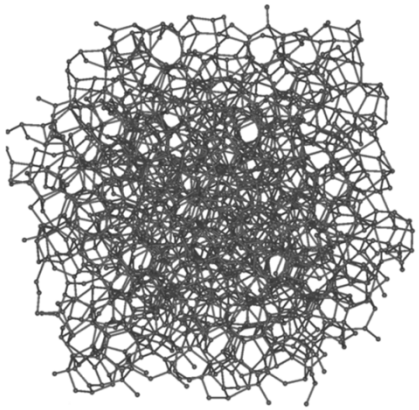
- Elements in group 3
- 3 valence electrons
- Exists as both non-metals and metals
- Metallic character increases down the group
- Aluminium (Al) used to produce many products e.g. cans, car body parts etc.
- Gallium (Ga) used in computer chips

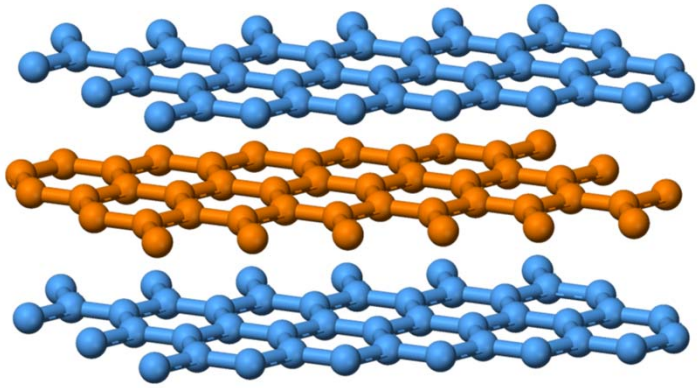
Carbon Family

- Elements in group 4
- 4 valence electrons
- Exists as both non-metals and metals
- Metallic character increases down the group
- Carbon (C) important element for all living organisms, form basis of branch of organic chemistry
- Silicon(Si) is a metalloid and is also abundant e.g. sand
- Germanium (Ge) is a metalloid, used in electronics as semi-conductors
- Tin (Sn) and Lead are metals, with high densities
- Leas (Pb) used in nuclear reactors and protection against radioactive materials

Allotropes of C

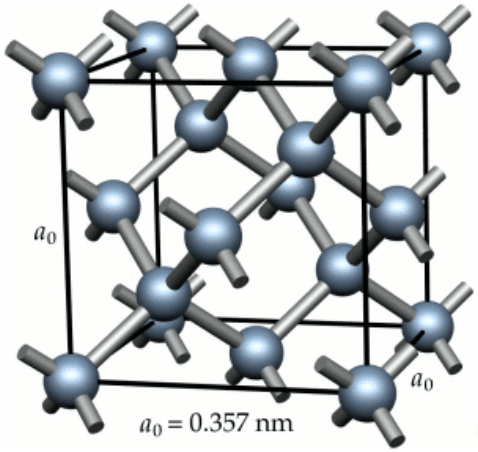
Amorphous C





Graphite
(Sheets)

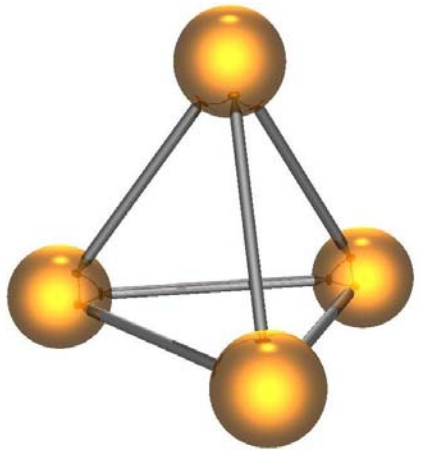
Diamond
(Network Solid)



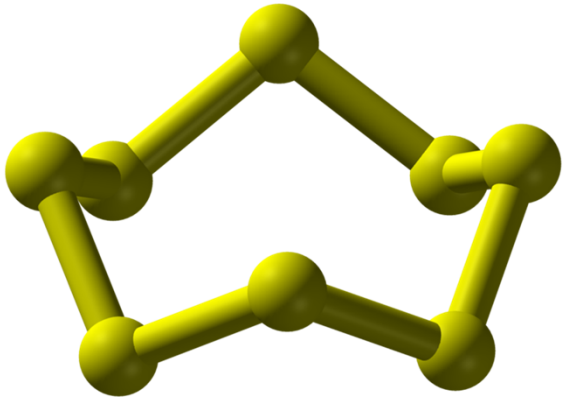


Nitrogen Family

- Elements in group 5
- 5 valence electrons
- Nitrogen makes up over ¾ of the atmosphere
- N_2 is exist in diatomic state
- Phosphorous (P) exist in two forms – white and red
- P used in explosives and in the manufacture of fertilizers



Oxygen (Chalcogens) Family



- Elements in group 6
- 6 valence electrons
- Tend to gain 2 electrons (form -2 ions)
- Oxygen(O) is a diatomic gas, essential for life
- Sulfur(S) is a solid, yellow non-metal – used in the manufacture of various chemical products e.g. sulfuric acid, paints etc.
- Selenium (Se) is a metal – good conductor and light sensitive, used in solar cells and photocopy machines
- Tellurium (Te) and Polonium (Po) are metalloids – radioactive

Halogens Family

- Elements in group 7
- 7 valence electrons
- Most reactive non-metal elements, tend to gain 1 electron (form -1 ions)
- Found in combination with other elements in nature e.g. NaCl
- Used to manufacture different products – disinfectants, bleach, plastic etc.
- Flourine (F) and Chlorine (Cl) are gases, Bromine (Br) is liquid, Iodine (I) and Astatine (At) are solid

Noble/Inert Gases

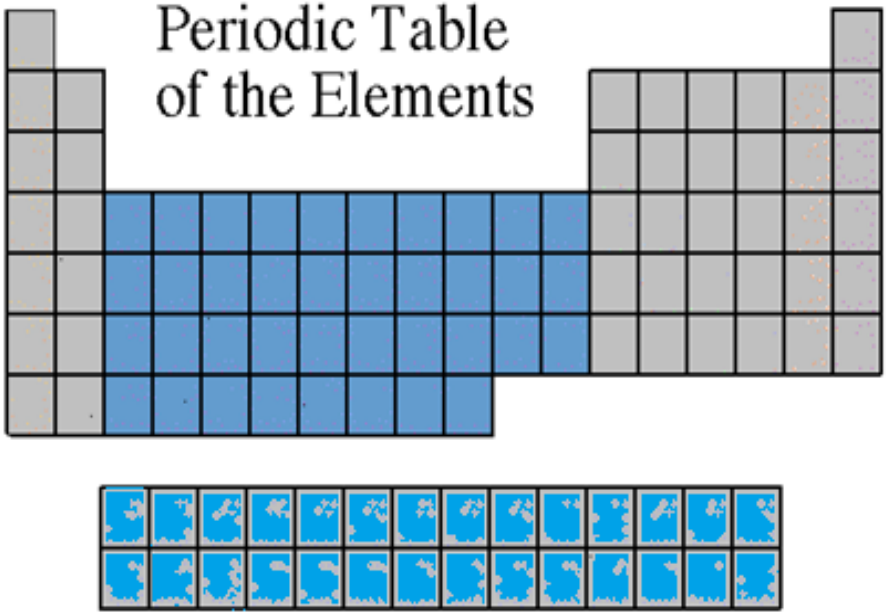
- Elements in group 8
- Have filled valence shell, overall charge is zero
- Unreactive (inert), monoatomic gases
- All are colourless, tasteless and odourless
- Helium (He) is less dense than air, used in balloons
- Neon (Ne) used in advertising lights, glows
- Radon (Rn) is radioactive



Transition Metals

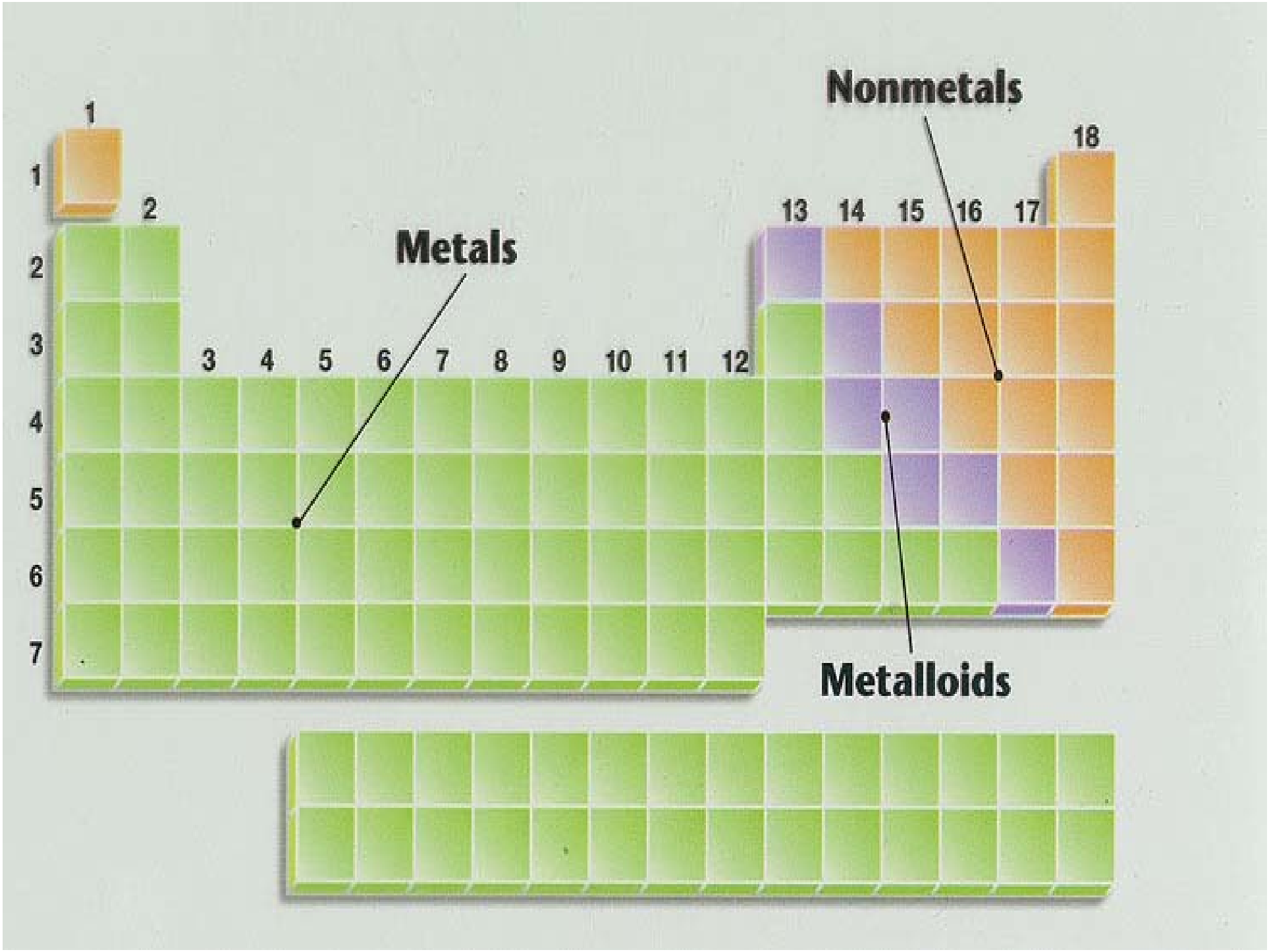
- Include elements in group B
- Arrangement of e- in outer shell vary, so the ion charge changes
- Lose diff # of valence e-, depending on the reaction
- Therefore, degree of reactivity and properties varies by element
- Two categories:
  - Main transition metals
    - e.g. Cu, Sn, Fe, Au, Ag
  - Inner transition metals
    - Lanthanides and Actinides

➢ Know the uses of some of the common metals



Division of the Periodic Table

- Different types of elements are found on different parts of the table
- 3 main classification:
  - **Metals** = to the left (majority of the elements).
  - **Nonmetals** = to the right (18 elements).
  - **Metalloids** = found on a "staircase" dividing metals and nonmetals (7 elements).
    - Lanthanides & Actinides (metals) added to bottom to make table manageable.

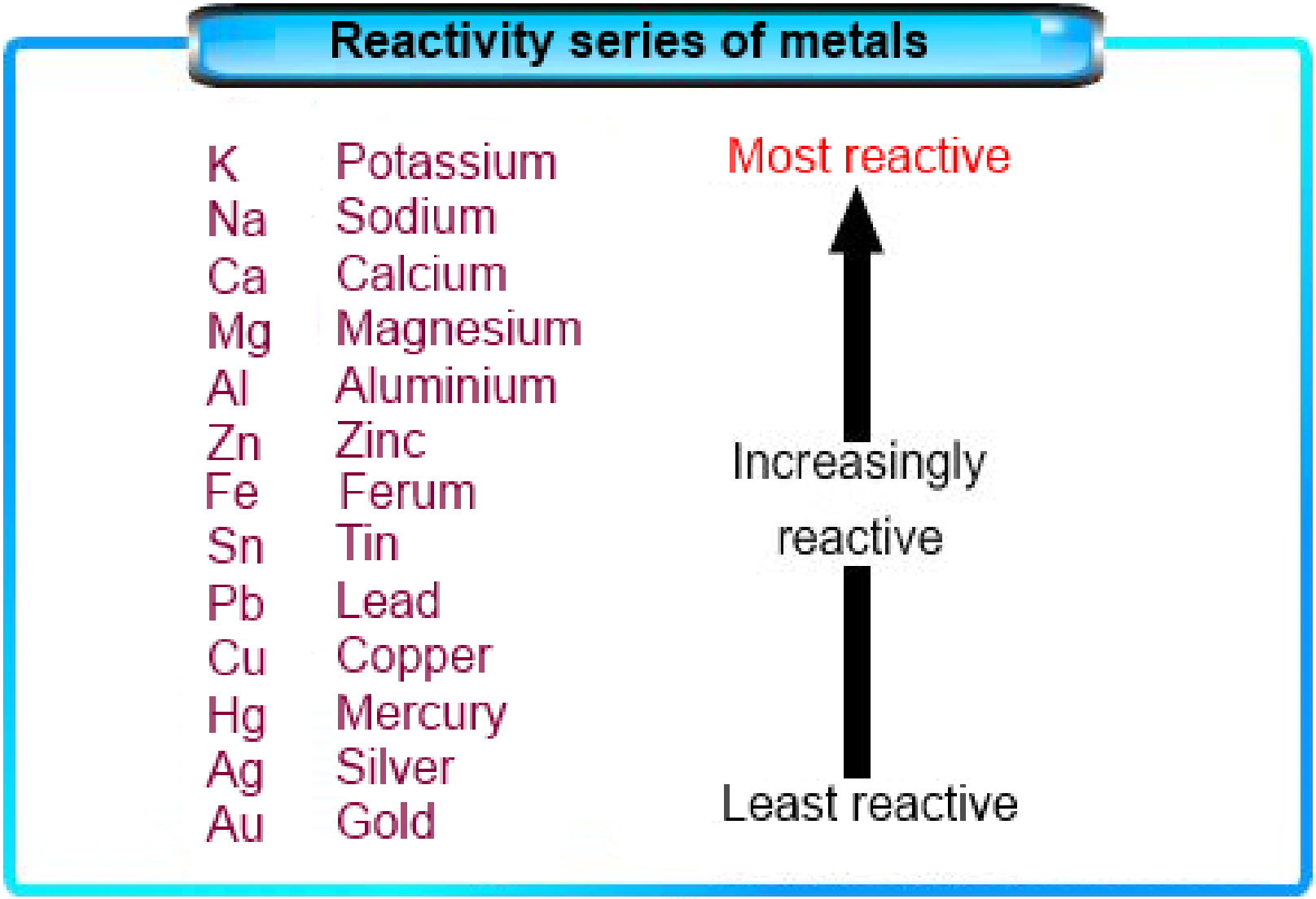

Nonmetals
Metals
Metalloids
1
2
3
4
5
6
7
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18

Metals

- Lustrous (shiny)
- Malleable (can be hammered into thin sheets)
- Ductile (can be pulled into wires)
- Conductive
  - Heat and electricity
- Solids (except mercury)
- High density, Mp and Bp
- React with O_2 to form oxides
- React with H_2O to form metal hydroxides
- React with acids to form Hydrogen gas.
- Uses: building structures, electric cables, radiators, colored paints, catalysts for industrial reactions, etc.



Reactivity of Metals



Minerals and Ores

- Unreactive metals are found in nature in their elemental state i.e. free
- Most metals found in nature in the form of minerals and ores
- Minerals – naturally occurring inorganic solids with a definite crystal structure
- Ores – concentrations of minerals in rock, that are high enough to be extracted (mining) for economic use
  - All ores are minerals, but not all minerals are ores





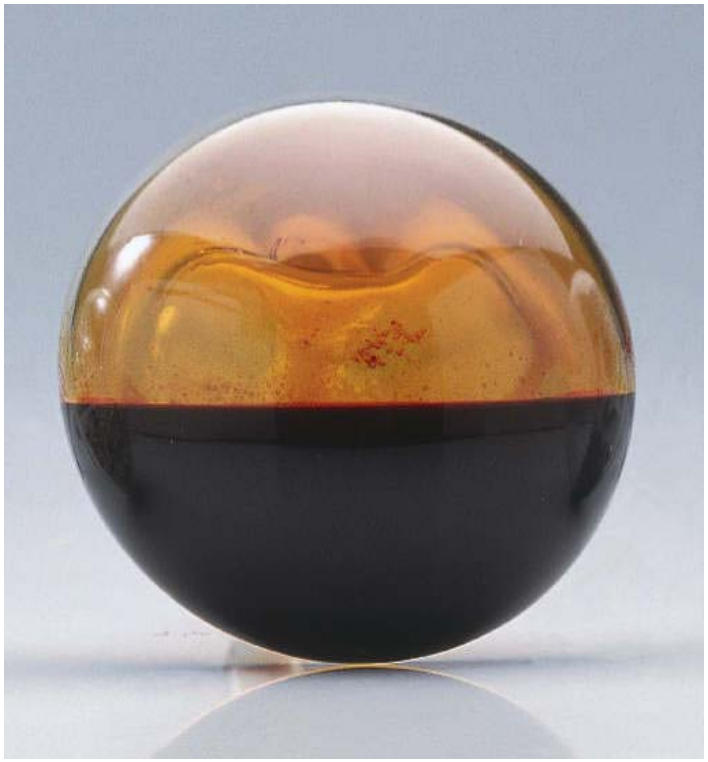
Alloys

- Mixture of two or more metal elements
- Mixture of alloy has different properties from those of the component elements
- Property is dependant on the types and amount of individual metals used.
- Common alloys include:
  - Sterling silver (Ag, Cu)
  - Brass (Zn, Cu)
  - Stainless Steel (Fe, Cr, Ni)
  - Duralium (Al, Cu)



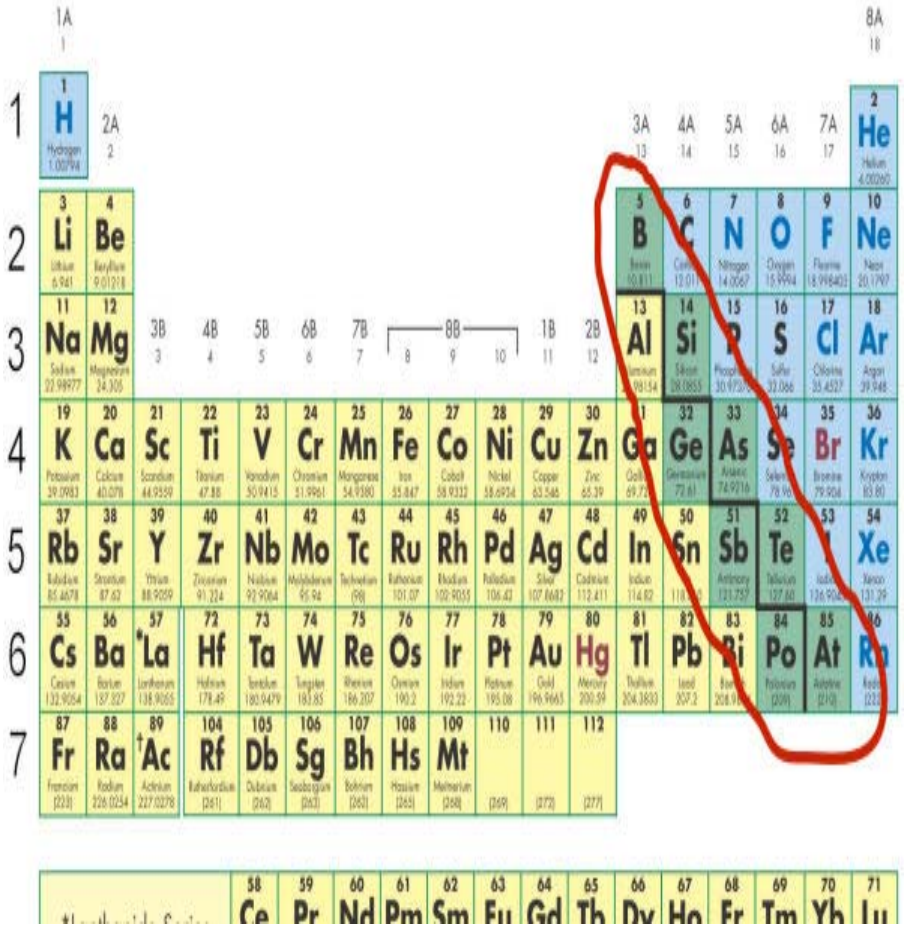
Non-metals

- Wide range of properties, opposite to that of metals
- Tend to:
  - Be Dull
  - Poor conductors
  - Gain e- during reactions
  - Many are gases at room temp
  - Some are brittle solids e.g. sulfur
  - Bromine = only non-metal which is liquid at room temp
  - Not react with acids
  - Have lower melting & boiling points.



Metalloids

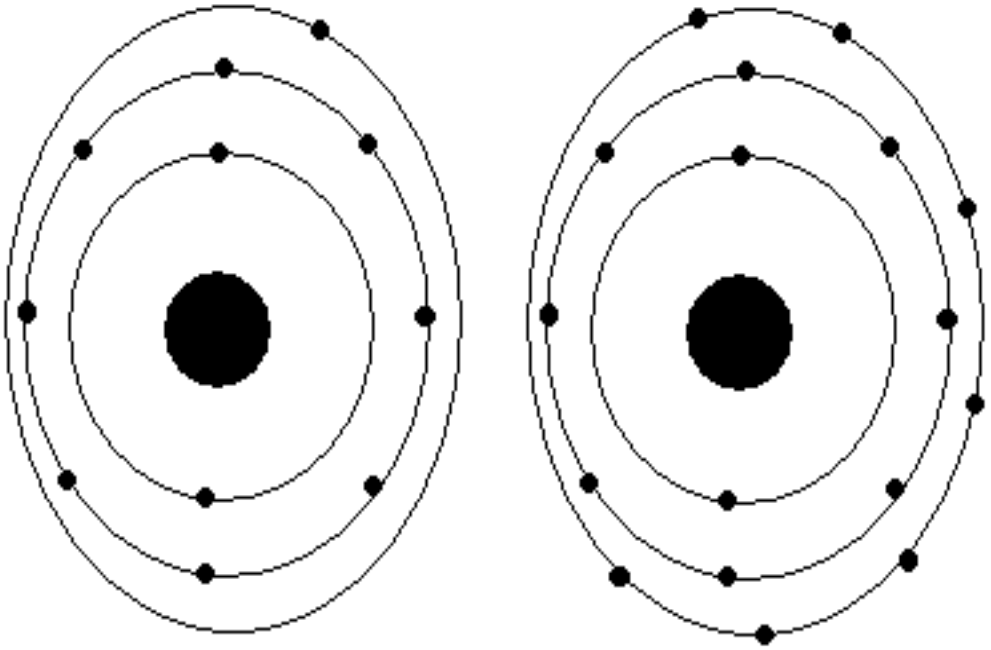
- Also called “semi-metals” or “staircase elements.”
- Combination of properties of metals and nonmetals.
- Boron, Silicon, Germanium, Arsenic, Antimony, Tellurium, & Polonium
- Many exhibit semi-conducting behavior.



THE END

Sodium

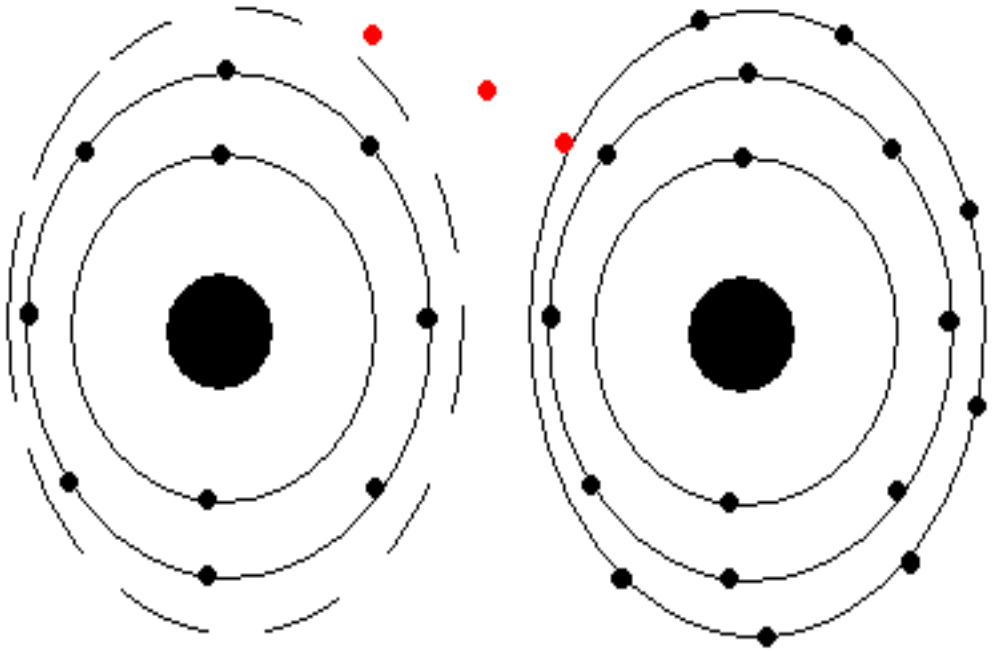
Chlorine



1 valence electron

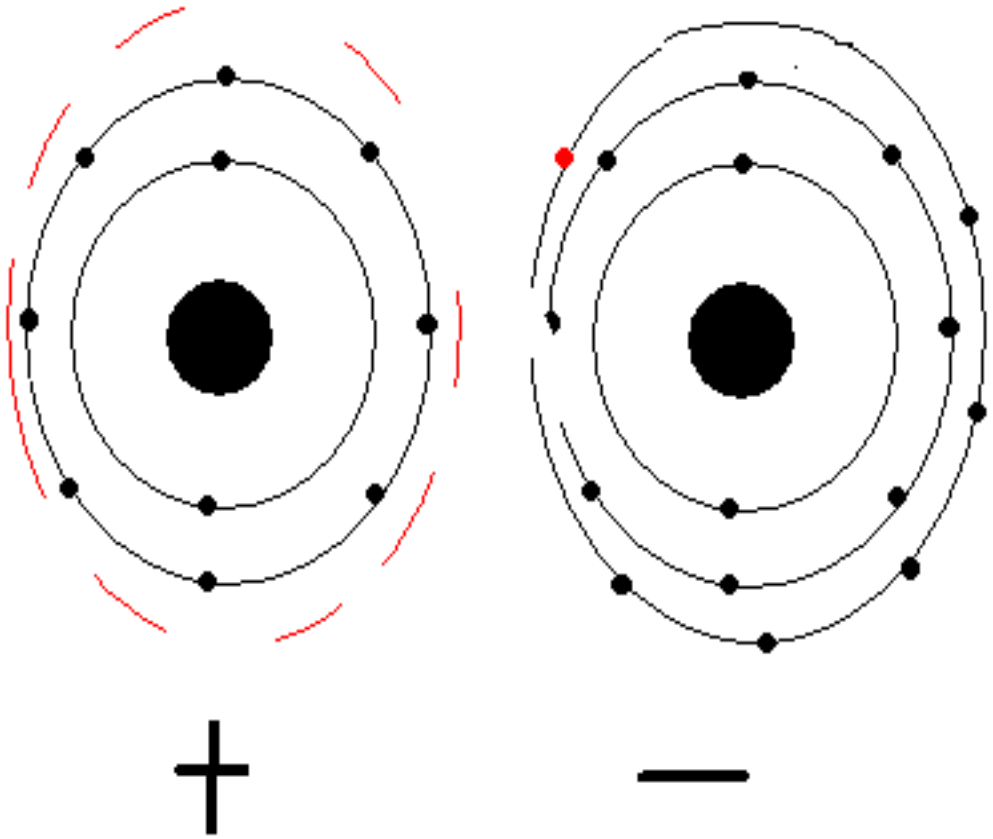
7 valence electrons

Sodium Chlorine

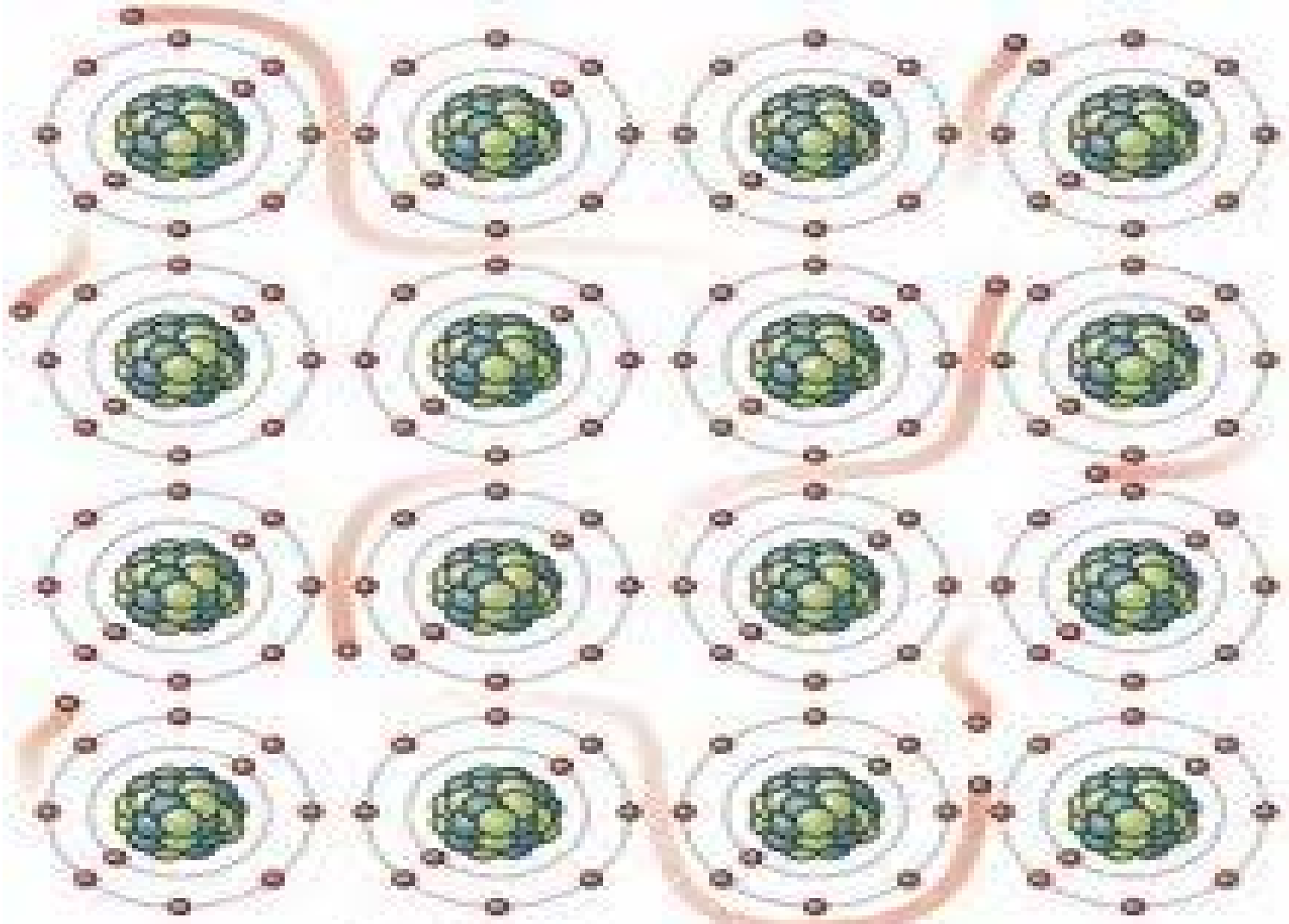


Sodium loses one electron.
Chlorine gains one electron.

Sodium Chloride



Metallic Bonding



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PERIODIC TABLE OF THE ELEMENTS



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PERIOD	GROUP 1 IA	2 IIA	3 IIIB	4 IVB	5 VB	6 VIB	7 VIIB	8	9 VIII B	10	11 IB	12 IIB	13 IIIA	14 IVA	15 VA	16 VIA	17 VIIA	18 VIIIA
1	1 1.0079 H HYDROGEN																	2 4.0026 He HELIUM
2	3 6.941 Li LITHIUM	4 9.0122 Be BERYLLIUM											5 10.811 B BORON	6 12.011 C CARBON	7 14.007 N NITROGEN	8 15.999 O OXYGEN	9 18.998 F FLUORINE	10 20.180 Ne NEON
3	11 22.990 Na SODIUM	12 24.305 Mg MAGNESIUM											13 26.982 Al ALUMINIUM	14 28.086 Si SILICON	15 30.974 P PHOSPHORUS	16 32.065 S SULPHUR	17 35.453 Cl CHLORINE	18 39.948 Ar ARGON
4	19 39.098 K POTASSIUM	20 40.078 Ca CALCIUM	21 44.956 Sc SCANDIUM	22 47.867 Ti TITANIUM	23 50.942 V VANADIUM	24 51.996 Cr CHROMIUM	25 54.938 Mn MANGANESE	26 55.845 Fe IRON	27 58.933 Co COBALT	28 58.693 Ni NICKEL	29 63.546 Cu COPPER	30 65.38 Zn ZINC	31 69.723 Ga GALLIUM	32 72.64 Ge GERMANIUM	33 74.922 As ARSENIC	34 78.96 Se SELENIUM	35 79.904 Br BROMINE	36 83.798 Kr KRYPTON
5	37 85.468 Rb RUBIDIUM	38 87.62 Sr STRONTIUM	39 88.906 Y YTTRIUM	40 91.224 Zr ZIRCONIUM	41 92.906 Nb NIOBIUM	42 95.94 Mo MOLYBDENUM	43 (98) Tc TECHNETIUM	44 101.07 Ru RUTHENIUM	45 102.91 Rh RHODIUM	46 106.42 Pd PALLADIUM	47 107.87 Ag SILVER	48 112.41 Cd CADMIUM	49 114.82 In INDIUM	50 118.71 Sn TIN	51 121.76 Sb ANTIMONY	52 127.60 Te TELLURIUM	53 126.90 I IODINE	54 131.29 Xe XENON
6	55 132.91 Cs CAESIUM	56 137.33 Ba BARIUM	57-71 La-Lu Lanthanide	72 178.49 Hf HAFNIUM	73 180.95 Ta TANTALUM	74 183.84 W TUNGSTEN	75 186.21 Re RHENIUM	76 190.23 Os OSMIUM	77 192.22 Ir IRIDIUM	78 195.08 Pt PLATINUM	79 196.97 Au GOLD	80 200.59 Hg MERCURY	81 204.38 Tl THALLIUM	82 207.2 Pb LEAD	83 208.98 Bi BISMUTH	84 (209) Po POLONIUM	85 (210) At ASTATINE	86 (222) Rn RADON
7	87 (223) Fr FRANCIUM	88 (226) Ra RADIUM	89-103 Ac-Lr Actinide	104 (267) Rf RUTHERFORDIUM	105 (268) Db DUBNIUM	106 (271) Sg SEABORGIUM	107 (272) Bh BOHRIUM	108 (277) Hs HASSIUM	109 (276) Mt MEITNERIUM	110 (281) Ds DARMSTADTIUM	111 (280) Rg ROENTGENIUM	112 (285) Cn COPERNICIUM						

LANTHANIDE	57 138.91 La LANTHANUM	58 140.12 Ce CERIUM	59 140.91 Pr PRASEODYMIUM	60 144.24 Nd NEODYMIUM	61 (145) Pm PROMETHIUM	62 150.36 Sm SAMARIUM	63 151.96 Eu EUROPIUM	64 157.25 Gd GADOLINIUM	65 158.93 Tb TERBIUM	66 162.50 Dy DYSPROSIUM	67 164.93 Ho HOLMIUM	68 167.26 Er ERBIUM	69 168.93 Tm THULIUM	70 173.05 Yb YTTERBIUM	71 174.97 Lu LUTETIUM
ACTINIDE	89 (227) Ac ACTINIUM	90 232.04 Th THORIUM	91 231.04 Pa PROTACTINIUM	92 238.03 U URANIUM	93 (237) Np NEPTUNIUM	94 (244) Pu PLUTONIUM	95 (243) Am AMERICIUM	96 (247) Cm CURIUM	97 (247) Bk BERKELIUM	98 (251) Cf CALIFORNIUM	99 (252) Es EINSTEINIUM	100 (257) Fm FERMIUM	101 (258) Md MENDELEVIUM	102 (259) No NOBELIUM	103 (262) Lr LAWRENCIUM

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